

Pranaya Pratik Das, Ph.D.

Theoretical Physicist, India
Quantum Chaos | Nonlinear Dynamics

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PROFESSIONAL SUMMARY

Theoretical physicist specialising in quantum chaos, nonlinear dynamics, and quantum information, with expertise in out-of-time-order correlators (OTOCs), spectral complexity, scrambling diagnostics, quantum scars, Lyapunov exponent analysis, and many-body dynamics. Experienced in analytical modelling, numerical simulation, Mathematica-based scientific computing, and cloud-based computational workflows. Research background spans black hole physics, active matter, statistical mechanics, nonlinear dynamical systems, and computational physics.

TECHNICAL SKILLS

Programming Languages: **Wolfram Mathematica, Fortran, MATLAB, Python**
Cloud Workflows: **Google Cloud Compute Engine VMs, Linux-based VM setup, SSH key authentication, PuTTY,**
Numerical Methods: **Multi-kernel parallelisation, matrix diagonalisation, numerical modelling, high-precision computation, data visualisation**
Web, Documentation, and Version Control: **HTML, CSS, Astro, GitHub, GitHub Pages, LaTeX**

RESEARCH INTERESTS

Classical and quantum chaos; nonlinear dynamics; information scrambling; OTOC diagnostics; quantum scars; spectral statistics; Lyapunov exponent analysis; black hole thermodynamics; active matter; computational physics

EDUCATION

Ph.D. in Theoretical Physics: Quantum Chaos <i>National Institute of Technology Rourkela</i> <i>Thesis Title: Diagnosis of Quantum Chaos in Perturbed Quantum Wells and Billiards</i> <i>Supervisor: Prof. Biplab Ganguli</i>	EQF: Level 8 <i>July 2019 – Dec 2025</i> <i>Rourkela, Odisha, India</i>
Master of Science: Physics <i>College of Basic Science and Humanities, OUAT</i> <i>Project Title: Dark Matter Signatures in Cosmic γ-Rays</i> <i>Supervisor: Prof. Manorama Panigrahy</i>	EQF: Level 7 <i>Aug 2014 – Aug 2016</i> <i>Bhubaneswar, Odisha, India</i>
Bachelor of Science: Physics <i>BJB Autonomous College, Utkal University</i> <i>Project Title: Black Hole and Its Mysteries</i>	EQF: Level 6 <i>July 2010 – May 2013</i> <i>Bhubaneswar, Odisha, India</i>

TEACHING EXPERIENCE

Teaching Assistance: B.Tech Physics Lab (Course Code: PH1070) • Instructor(s): Prof. Pitanmber Mahanandia, Prof. Susanta Kumar Bisoi, Prof. Indrani Banarjee, Prof. Biplab Ganguli	<i>NIT Rourkela, 2021–2023</i>
Teaching Assistance: U.G. Thermal Physics Lab (Course Code: PH3074) • Instructor(s): Prof. Biplab Ganguli	<i>NIT Rourkela, 2022–2023</i>
Teaching Assistant to Prof. Biplab Ganguli • Course(s): Nonlinear Dynamics, Chaos, and Its Recent Applications (PH6114); Advanced Statistical Mechanics (PH6112); Introduction to Classical Mechanics; Electrodynamics (PH4002)	<i>NIT Rourkela, 2021–2025</i>
M.Sc. Research Projects • Vipul Sharma Roll No.: 420PH2095 Thesis Title: Two & three scroll chaotic attractors 2022 • Ayush Sahu Roll No.: 418PH5033 Thesis Title: Nonlinear systems and chaos 2023 • Karishma Kujur Roll No.: 421PH2125 Thesis Title: Quantum chaos 2023 • Zubair Ahmad Kumar Roll No.: 422PH2069 Thesis Title: Diagnosing quantum chaos using OTOC 2024 • Vivek Sheoran Roll No.: 422PH2082 Thesis Title: Signature of chaos in quantum mechanics 2024	<i>NIT Rourkela, 2021–2024</i>

RESEARCH EXPERIENCE

- Deployed and managed Mathematica-based numerical simulations on Google Cloud Compute Engine VMs with 32 vCPUs, 128 GB RAM, and persistent disk storage, using Linux VM setup, SSH key-based authentication, and PuTTY/PuTTYgen access to perform Lyapunov exponent analysis, eigenvalue computations, and OTOC-based chaos diagnostics in quantum chaos and nonlinear dynamics research.
- Improved simulation throughput by parallelising Wolfram Language routines across multiple kernels, reducing computation time for large parameter scans and spectral calculations.
- Optimised cloud-based computational workflows by balancing VM performance, storage requirements, and cost efficiency for extended simulation runs.
- **Computational Research:** Numerical simulations, Lyapunov Exponent calculation, Poincaré Section, eigenvalue computations, spectral statistics, OTOC analysis, scrambling diagnostics, quantum chaos, nonlinear dynamics, data visualisation, reproducible computation

PUBLICATIONS

- | | |
|---|--|
| (1) Morphology-resolved scrambling in a chaotic quantum billiard
arXiv preprint
Author(s): Pranaya Pratik Das
DOI: https://doi.org/10.48550/arXiv.2606.16865 | June 2026
Under Review

ISSN: 2331-8422 |
| (2) Chaos in cymatics-inspired Gaussian landscapes
arXiv preprint
Author(s): Tanmayee Patra, Pranaya Pratik Das , Biplab Ganguli
DOI: https://doi.org/10.48550/arXiv.2606.09442 | June 2026
Under Review

ISSN: 2331-8422 |
| (3) Classical and quantum chaos in bean- and peanut-shaped billiards
arXiv preprint
Author(s): Pranaya Pratik Das , Tanmayee Patra, Biplab Ganguli
DOI: https://doi.org/10.48550/arXiv.2501.08839 | May 2026
Under Review

ISSN: 2331-8422 |
| (4) Cyclically symmetric Thomas oscillators as swarmalators: A model for active fluids and pattern formation
Journal: <i>Communications in Nonlinear Science and Numerical Simulation</i> , Elsevier (Q1)
Author(s): Vinesh Vijayan, Pranaya Pratik Das , K Hariprasad, P Sathish Kumar
DOI: https://doi.org/10.1016/j.cnsns.2025.109216 | Aug 2025

pp. 109216
Impact factor: 3.8
ISSN: 10075704 |
| (5) Out-of-Time-Order Correlation in perturbed quantum wells
Journal: <i>European Physical Journal D</i> , Springer Science (Q3)
Author(s): Pranaya Pratik Das , Biplab Ganguli
DOI: https://doi.org/10.1140/epjd/s10053-025-01025-7 | June 2025
Vol. 79 no. 6 pp. 74
Impact factor: 1.5
ISSN: 14346060, 14346079 |
| (6) Interplay between the Lyapunov exponents and phase transitions of charged AdS black holes
Journal: <i>Physical Review D</i> , American Physical Society (Q1)
Author(s): Bhaskar Shukla, Pranaya Pratik Das , David Dudal, Subhash Mahapatra
DOI: https://doi.org/10.1103/PhysRevD.110.024068 | July 2024

vol. 110 no. 2 pp. 024068
Impact factor: 5.3
ISSN: 24700010, 24700029 |
| (7) Dynamics of a charged Thomas oscillator in an external magnetic field
Journal: <i>Physica Scripta</i> , IOP Science (Q2)
Author(s): Vinesh Vijayan, Pranaya Pratik Das
DOI: 10.1088/1402-4896/ac99ab | Oct 2022
Vol. 97, no. 11
Impact factor: 2.6
ISSN: 00318949, 14024896 |

CONFERENCES, WORKSHOPS AND SCHOOLS

- | | |
|--|---|
| (1) 60 Years of DFT: Advancements in Theory & Computation
<i>Poster Presentation</i> | 2024
<i>IIT Mandi, India</i> |
| (2) HPC Symposium
<i>Poster Presentation</i> | 2024
<i>NIT Rourkela, India</i> |
| (3) Secrets of getting published in high impact factor journals
<i>Attended online</i> | 2023
<i>Wiley</i> |
| (4) Integrability, Deformations and Chaos
<i>Attended online</i> | 2023
<i>OIST, Onna, Okinawa, Japan</i> |

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|---|---------------------------------------|
| (5) School on Quantum Chaos
Attended online | 2023
ICTP-SAIFR, São Paulo, Brazil |
| (6) Complex Lagrangian Problems of Particles in Flows
Attended online | 2022
ICTS-TIFT, India |
| (7) Spring College in the Physics of Complex Systems
Attended online | 2022
ICTP, Trieste, Italy |

AWARDS, SCHOLARSHIPS, AND ACHIEVEMENTS

Qualified: National Graduate Physics Examination (NGPE) conducted by IAPT	2012
Scholarships: P.G. Meritorious Scholarship by Institute of Mathematics and Applications (IMA)	2014–2016
Scholarships: Medhabruti Scholarship by Odisha State Government	2014-2016
Qualified: TIFR GS-2016 in Physics	2016
Qualified: GATE in Physics with 94.7901 Percentile	2016

LANGUAGE SKILLS

Mother Tongue: Odia

Other Languages:

English:	Listening: C2, Spoken production: C2, Reading: C2, Spoken interaction: C1, Writing: C2
Hindi	Listening: C2, Spoken production: C2, Reading: B2, Spoken interaction: C2, Writing: B2
Sanskrit	Listening: B2, Spoken production: A1, Reading: B2, Spoken interaction: A1, Writing: B1
Levels: A1 & A2: Basic user; B1 & B2: Independent user; C1 & C2: Proficient user	

ONLINE PROFILES

Personal Homepage: <https://ppdws.github.io/pranayapratikdas/>
 Scopus ID: **57451177800**
 Google Scholar: **Pranaya Pratik Das**
 LinkedIn: **pranaya-pratik-das-28a94439b**

arXiv: **das_p_3**
 ORCID ID: **0000-0002-6025-7719**
 ResearchGate: **Pranaya-Das**
 Instagram: **pranaya.pratik.das**

REFERENCES

Prof. Biplab Ganguli Department of Physics and Astronomy Email: biplabg@nitrkl.ac.in or biplab62g@gmail.com Relationship: PhD supervisor	Professor NIT Rourkela
Prof. Subhash Chandra Mahapatra Department of Physics and Astronomy Email: mahapatrasub@nitrkl.ac.in or subhashmahapatra@gmail.com Relationship: Research collaborator and course instructor	Associate Professor NIT Rourkela
Prof. Mithun Biswas Department of Physics and Astronomy Email: biswasm@nitrkl.ac.in Relationship: PhD DSC member and course instructor	Assistant Professor NIT Rourkela
Prof. Chhatrapati Parida Department of Physics Email: cparida@ouat.ac.in Relationship: Course instructor	Professor College of Basic Science and Humanities, OUAT

Pranaya Pratik Das
June 17, 2026